

Prayas JEE 2025

Practice Sheet (Mathematics)

Quadratic Equations

- Q1** The number of roots of the equation $\sqrt{x-2}(x^2-4x+3)=0$ is
 (A) Three (B) Four
 (C) One (D) Two
- Q2** The sum of all the real values of x satisfying the equation $2^{(x-1)(x^2+5x-50)} = 1$ is:
 (A) 16 (B) 14
 (C) -4 (D) -5
- Q3** If both the roots of the quadratic equation $x^2 - 2px + p^2 + p - 5 = 0$ are less than 3, then k lies in the interval
 (A) $(-\infty, 1)$
 (B) $(1, \infty)$
 (C) $(1, 3)$
 (D) $(3, \infty)$
- Q4** For what values of the parameter k in the inequality $\left| \frac{x^2+kx+1}{x^2+x+1} \right| < 3$, satisfied
 (A) $-1 < k < 5$
 (B) $1 < k < 5$
 (C) Either
 (D) None
- Q5** If the quadratic equations $ax^2 + 2cx + b = 0$ and $ax^2 + 2bx + c = 0 (b \neq c)$ have a common root, then $a + 4b + 4c$ is equal to
 (A) -2 (B) -1
 (C) 0 (D) 1
- Q6** If α, β are real and $\alpha^2, -\beta^2$ are the roots of $a^2x^2 + x + 1 - a^2 = 0; (a > 1)$, then $\beta^2 =$
 (A) a^2 (B) 1
 (C) $1 - a^2$ (D) $1 + a^2$
- Q7** The equation $\sin^4 x - 2\cos^2 x + a^2 = 0$ is solvable if
 (A) $\sqrt{3} \geq a \geq -\sqrt{3}$
 (B) $\sqrt{2} \geq a \geq -\sqrt{2}$
 (C) $1 \geq a \geq -1$
 (D) None of these
- Q8** The value of a for which the quadratic equation $3x^2 + 2a^2x + a^2 - 3a + 2 = 0$ possesses roots of opposite signs lies in
 (A) $(-\infty, 1)$
 (B) $(-\infty, 0)$
 (C) $(1, 2)$
 (D) $(\frac{3}{2}, 2)$
- Q9** If α and $\beta (\alpha < \beta)$, are the roots of the equation $x^2 + bx + c = 0$, where $c < 0 < b$, then
 (A) $0 < \alpha < \beta$
 (B) $\alpha < 0 < \beta < |\alpha|$
 (C) $\alpha < \beta < 0$
 (D) $\alpha < 0 < |\alpha| < \beta$
- Q10** If the roots of the equation $x^3 + 3px^2 + 3qx + r = 0$ are in AP, then the condition is:
 (A) $2p^3 = 3pq + r$
 (B) $2p^3 = 3pq$
 (C) $2p^3 + r = 3pq$
 (D) $2p^3 - r = -3pq$
- Q11** If the equation $x^4 - 4x^3 + ax^2 + bx + 1 = 0$ has four positive roots, then the value of $(a + b)$ is:
 (A) -4
 (B) 2
 (C) 6
 (D) Cannot be determined
- Q12**



If α, β be the roots of

$$4x^2 - 17x + \lambda = 0, \lambda \in R \text{ such that}$$

$1 < \alpha < 2$ and $2 < \beta < 3$, then the number of integral values of λ is

- (A) 1 (B) 2
(C) 3 (D) 4

Q13 The set of all real numbers x for which

$$x^2 - |x + 2| + x > 0 \text{ is}$$

- (A) $(-\infty, -2) \cup (2, \infty)$
(B) $(-\infty, -\sqrt{2}) \cup (\sqrt{2}, \infty)$
(C) $(-\infty, -1) \cup (1, \infty)$
(D) $(\sqrt{2}, \infty)$

Q14 If λ_1 and λ_2 are the two values of λ such that the roots α and β of the quadratic equation, $\lambda(x^2 - x) + x + 5 = 0$

satisfy $\frac{\alpha}{\beta} + \frac{\beta}{\alpha} + \frac{4}{5} = 0$, then $\frac{\lambda_1}{\lambda_2} + \frac{\lambda_2}{\lambda_1}$ equal to:

- (A) 536 (B) 512
(C) 504 (D) 488

Q15

	Column - I		Column - II
I.	Ramesh and Mahesh solve an equation. In solving Ramesh commits a mistake in constant term and finds the root 8 and 2. Mahesh commits a mistake in the coefficient of x and finds the roots -9 and -1 . Find the sum of correct roots.	P.	7
II.	α & β are the roots of the equation $x^2 - 7x - 1 = 0$, then $\frac{\alpha^{10} - \beta^{10} - (\alpha^8 - \beta^8)}{\alpha^{\alpha+\beta+2} - \beta^{\alpha+\beta+2}}$	Q.	114
III.	The minimum value of $(x-6)^2 + (x+3)^2 + (x-8)^2 + (x+4)^2 + (x-3)^2$ is	R.	10
IV.	The number of integral values of k for which the equation $3 \sin x + 4 \cos x = k + 1$ has a solution, $k \in R$ is	S.	11

- (A) I-P, II-S, III-R, IV-Q
(B) I-S, II-R, III-P, IV-Q
(C) I-R, II-P, III-Q, IV-S
(D) I-R, II-P, III-S, IV-Q

Q16 The number of real solutions of the equation $|x|^2 - 3|x| + 2 = 0$ is:

Q17 Let $\alpha, \beta (\alpha > \beta)$ be the roots of the quadratic equation $x^2 - x - 4 = 0$. If $P_n = \alpha^n - \beta^n$, $n \in N$, then $(P_{15}P_{16} - P_{14}P_{16} - P_{15}^2 + P_{14}P_{15}) / P_{13}P_{14}$ is equal to

Q18 If the equation $ax^2 + bx + c = 0 (a > 0)$ has two roots α and β such that $\alpha < -2$ and $\beta > 2$, then

- (A) $b^2 - 4ac > 0$
(B) $4a + 2|b| + c < 0$
(C) $a + |b| + c < 0$
(D) All of these

Q19 For each positive number x , let

$$f(x) = \frac{(x + \frac{1}{x})^6 - (x^6 + \frac{1}{x^6}) - 2}{(x + \frac{1}{x})^3 + (x^3 + \frac{1}{x^3})}$$

of $f(x)$ is

- (A) 1 (B) 2
(C) 3 (D) 6

Q20 If α, β, γ are roots of $x^3 + px + r = 0$ then

$$\frac{1+\alpha}{1-\alpha} + \frac{1+\beta}{1-\beta} + \frac{1+\gamma}{1-\gamma} =$$

- (A) $\frac{3-p-3r}{1+p+r}$
(B) $\frac{3+p+3r}{1+p+r}$
(C) $\frac{3+r+p}{1+p+r}$
(D) 1

Q21 Let x_1, x_2, x_3, x_4 be the roots (real or complex) of the equation $x^4 + ax^3 + bx^2 + cx + d = 0$. If $x_1 + x_2 = x_3 + x_4$ and $a, b, c, d \in R$, then If $a = 2$, then the value of $b - c$ is

- (A) -1 (B) 1
(C) -2 (D) 2

Q22 Consider the equation $x^2 + 2x - n = 0$, where $n \in N$ and $n \in [5, 100]$. Total number of different values of n so that the given equation has integral roots is

- (A) 8 (B) 3
(C) 6 (D) 4



Q23 Consider the equation $x^3 - ax^2 + bx - c = 0$, where $a, b, c \in \mathbb{Q}; (a \neq 1)$. It is given that x_1, x_2 and x_1x_2 are the real roots of the equation. If $b + c = 2(a + 1)$, then $x_1x_2 \left(\frac{a+1}{b+c}\right)$ is equal to.

Q24 If α, β are the roots of the equation $2x^2 - 35x + 2 = 0$ then the value of $(2\alpha - 35)^3(2\beta - 35)^3$ is equal to
(A) 8 (B) 1
(C) 64 (D) None of these

Q25 Let α, β be the roots of $ax^2 + bx + c = 0$ and γ, δ be the roots of $px^2 + qx + r = 0$; and D_1, D_2 be the respective discriminants of these equations. If $\alpha, \beta, \gamma, \delta$ are in AP , then $D_1 : D_2$ equal to
(A) $\frac{a^2}{p^2}$
(B) $\frac{a^2}{b^2}$
(C) $\frac{b^2}{q^2}$
(D) $\frac{c^2}{r^2}$

Q26 If α, β, γ are the roots of the equation $x^3 + 4x + 1 = 0$, then find the value of $(\alpha + \beta)^{-1} + (\beta + \gamma)^{-1} + (\gamma + \alpha)^{-1}$

Q27 If α, β be roots of $x^2 + px + 1 = 0$ and γ, δ be the roots of $x^2 + qx + 1 = 0$, then $(\alpha - \gamma)(\beta - \gamma)(\alpha + \delta)(\beta + \delta)$ is equal to
(A) $p^2 + q^2$
(B) $p^2 - q^2$
(C) $q^2 - p^2$
(D) None of these

Q28 The number of ordered pairs (a, b) , where a, b are integers satisfying the inequality $\min(x^2 + (a - b)x + (1 - a - b)) > \dots$, is:
 $\max(-x^2 + (a + b)x - (1 + a + b)) \forall x \in \mathbb{R}$

Q29 Let α, β be two real numbers satisfying the following relations
 $\alpha^2 + \beta^2 = 5, 3(\alpha^5 + \beta^5) = 11(\alpha^3 + \beta^3)$

Possible value of $\alpha + \beta$

(A) ± 2
(B) ± 3
(C) ± 1
(D) $\pm \sqrt{3}$

Q30 If the equation $2^{2x} + a \cdot 2^{x+1} + a + 1 = 0$ has roots of opposite sign, then the exhaustive set of real values of a is

(A) $(-\infty, 0)$
(B) $\left(-1, -\frac{2}{3}\right)$
(C) $\left(-\infty, -\frac{2}{3}\right)$
(D) $(-1, \infty)$

Q31 If α, β and γ are roots of $4x^3 + 8x^2 - x - 2 = 0$, then the value of $\frac{4(\alpha+1)(\beta+1)(\gamma+1)}{\alpha\beta\gamma}$ is equal to

(A) $\frac{3}{2}$
(B) 6
(C) $-\frac{3}{2}$
(D) -6

Q32 The equations $x^3 + 5x^2 + px + q = 0$ and $x^3 + 7x^2 + px + r = 0$ have two roots in common. If the third root of each equation is represented by x_1 and x_2 , respectively, then the ordered pair (x_1, x_2) is

(A) $(-5, -7)$ (B) $(1, -1)$
(C) $(-1, 1)$ (D) $(5, 7)$

Q33 If $x^3 - x^2 + 5x - 1 = 0$ has roots α, β, γ and $x^3 + ax^2 + bx + c = 0$ has roots $\alpha\beta, \beta\gamma, \gamma\alpha$ then $(a + b + c)$ is

(A) -1 (B) -5
(C) 3 (D) 7

Q34 If α, β are the roots of equation $x^2 + x - 1 = 0$, then the value of $\frac{\alpha\beta^4(\beta+1)^4 + \beta\alpha^4(\alpha+1)^4}{\alpha^2 + \beta^2 + \alpha + \beta}$ is equal to:

(A) $\frac{1}{2}$
(B) $-\frac{1}{2}$
(C) 2
(D) -2



Q35 If x_1, x_2, x_3 are such that

$$\begin{aligned}x_1 + x_2 + x_3 &= 2 \\x_1^2 + x_2^2 + x_3^2 &= 6 \\x_1^3 + x_2^3 + x_3^3 &= 8\end{aligned}$$

then value of $-(x_2 - x_3)(x_3 - x_1)(x_1 - x_2)$ is

Q36 Let $f(x)$ be a polynomial of degree 5 with leading coefficient unity such that $f(1) = 5, f(2) = 4, f(3) = 3, f(4) = 2, f(5) = 1$. Then $f(6)$ is equal to :

- (A) 0 (B) 24
(C) 120 (D) 720

Q37 If $\alpha, \beta, \gamma, \delta \in R$ satisfy $\frac{(\alpha+1)^2 + (\beta+1)^2 + (\gamma+1)^2 + (\delta+1)^2}{\alpha + \beta + \gamma + \delta} = 4$

if biquadratic equation $a_0x^4 + a_1x^3 + a_2x^2 + a_3x + a_4 = 0$ has the roots

$$\left(\alpha + \frac{1}{\beta} - 1\right), \left(\beta + \frac{1}{\gamma} - 1\right), \left(\gamma + \frac{1}{\delta} - 1\right), \left(\delta + \frac{1}{\alpha} - 1\right)$$

Then the value of $\frac{a_2}{a_0}$ is

- (A) 4 (B) -4
(C) 6 (D) None of these

Q38 Let α, β be the roots of the equation $x^2 - \sqrt{2}x + 2 = 0$.

Then $\alpha^{14} + \beta^{14}$ is equal to

- (A) $-64\sqrt{2}$ (B) $-128\sqrt{2}$
(C) -64 (D) -128

Q39 Let $f(x)$ be a quadratic polynomial such that $f(-2) + f(3) = 0$.

If one of the roots of $f(x) = 0$ is -1, then the sum of the roots

of $f(x) = 0$ is equal to:

- (A) $\frac{11}{3}$ (B) $\frac{7}{3}$
(C) $\frac{13}{3}$ (D) $\frac{14}{3}$

Q40

If α, β are the roots of the equation $x^2 - \left(5 + 3\sqrt{\log_3 5} - 5\sqrt{\log_5 3}\right)x + 3\left(3^{(\log_3 5)^{\frac{1}{3}}} - 5^{(\log_5 3)^{\frac{2}{3}}} - 1\right) = 0$

then the equation, whose roots are $\alpha + \frac{1}{\beta}$ and $\beta + \frac{1}{\alpha}$,

- (A) $3x^2 - 20x - 12 = 0$ (B) $3x^2 - 10x - 4 = 0$
(C) $3x^2 - 10x + 2 = 0$ (D) $3x^2 - 20x + 16 = 0$

Q41 If α and β be two roots of the equation $x^2 - 64x + 256 = 0$. Then the value of $\left(\frac{\alpha^3}{\beta^5}\right)^{\frac{1}{8}} + \left(\frac{\beta^3}{\alpha^5}\right)^{\frac{1}{8}}$ is

- (A) 3 (B) 2
(C) 4 (D) 1

Q42 Let α and β be the roots of the equation $x^2 - x - 1 = 0$. If $p_k = (\alpha)^k + (\beta)^k, k \geq 1$, then which of the following statements is not true?

- (A) $p_5 = p_2 \cdot p_3$
(B) $(p_1 + p_2 + p_3 + p_4 + p_5) = 26$
(C) $p_3 = p_5 - p_4$
(D) $p_5 = 11$

Q43 Let S be the set of all real roots of the equation, $3^x(3^x - 1) + 2 = |3^x - 1| + |3^x - 2|$. Then S :

- (A) contains exactly two elements
(B) is a singleton
(C) contains at least four elements
(D) is an empty set

Q44 Consider the quadratic equation $(c - 5)x^2 - 2cx + (c - 4) = 0, c \neq 5$. Let S be the set of all integral values of c for which one root of the equation lies in the interval $(0, 2)$ and its other root lies in the interval $(2, 3)$. Then the number of elements in S is:

- (A) 18 (B) 12
(C) 10 (D) 11

Q45



$$\text{Let } S = \left\{ x : x \in R \text{ and } (\sqrt{3} + \sqrt{2})^{x^2-4} + (\sqrt{3} - \sqrt{2})^{x^2-4} = 10 \right\}$$

Then $n(S)$ is equal to

- (A) 2 (B) 4
(C) 6 (D) 0

Q46 The equation $x^2 - 4x + [x] + 3 = x[x]$, where $[x]$ denotes the greatest integer function, has:

- (A) exactly two solutions in $(-\infty, \infty)$
(B) no solution
(C) a unique solution in $(-\infty, 1)$
(D) a unique solution in $(-\infty, \infty)$

Q47 Let p and q be two real numbers such that $p + q = 3$ and $p^4 + q^4 = 369$. Then $\left(\frac{1}{p} + \frac{1}{q}\right)^{-2}$ is equal to _____.

Q48 The number of real solutions of the equation $e^{4x} + 4e^{3x} - 58e^{2x} + 4e^x + 1 = 0$ is

Q49 The number of distinct real roots of the equation $3x^4 + 4x^3 - 12x^2 + 4 = 0$ is _____.

Q50 Let α and β be two real numbers such that $\alpha + \beta = 1$ and $\alpha\beta = -1$. Let $P_n = (\alpha)^n + (\beta)^n$, $P_{n-1} = 11$ and $P_{n+1} = 29$ for some integer $n \geq 1$. Then, the value of P_n^2 is _____



Answer Key

Q1 (D)
Q2 (C)
Q3 (A)
Q4 (A)
Q5 (C)
Q6 (B)
Q7 (B)
Q8 (C)
Q9 (B)
Q10 (C)
Q11 (B)
Q12 (B)
Q13 (B)
Q14 (D)
Q15 (C)
Q16 4
Q17 16
Q18 (D)
Q19 (D)
Q20 (A)
Q21 (B)
Q22 (A)
Q23 1
Q24 (C)
Q25 (A)

Q26 4
Q27 (C)
Q28 6
Q29 (B)
Q30 (B)
Q31 (D)
Q32 (A)
Q33 (B)
Q34 (B)
Q35 6
Q36 (C)
Q37 (C)
Q38 (D)
Q39 (A)
Q40 (B)
Q41 (B)
Q42 (A)
Q43 (B)
Q44 (D)
Q45 (B)
Q46 (D)
Q47 4
Q48 2
Q49 4
Q50 324



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